Curriculum Vitae – Nicholas Anton Collins-Craft

Education

Doctor of Philosophy/Doctorat from the University of Sydney and École nationale des ponts et chaussées (2015 – 2019).
Advisors: Jointly supervised under a cotutelle agreement by Professor Itai Einav (Sydney), Professeur Jean Sulem and Dr Ioannis Stefanou (ENPC).
Thesis title: *The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media.* Defended the 22nd of November, 2019.

Bachelor of Engineering in Civil Engineering with First Class Honours and the University Medal from the University of Sydney (2011 – 2014).
Honours Thesis advisor: Professor Itai Einav.
Thesis title: *The Contribution of Granular Rotation to Improving the Efficiency of Heat Transfer.*

Certificate of Advanced Engineering from the University of Sydney (2011 – 2014).
Undertaken as an extension programme in parallel with the Bachelor's Degree, enabling high-achieving students to work on projects in humanitarian engineering, educational outreach and product commercialisation.

Higher School Certificate at North Sydney Boys High School (2005 – 2010).

Academic Employment

May 2026 – Present Lecturer in Applied Mathematics at Murdoch University.

November 2024 – April 2026 Postdoctoral researcher in the geotechnical team of Laboratoire Navier at École nationale des ponts et chaussées, supervised by Professeur Jean Sulem.

September 2022 – October 2024 Marie Skłodowska-Curie Actions Postdoctoral Fellow in the Tripop team at the Inria Centre of the Université Grenoble Alpes, financed by the project LEMMA (Landslide and avalanchE Mechanics with Multiphysical datA), in collaboration with Dr Vincent Acary (Inria), Dr Franck Bourrier (Inrae & Inria) and Professor Johan Gaume (WSL & ETH Zürich).

November 2020 – August 2022 Postdoctoral researcher in the Tripop team at the Inria Centre of the Université Grenoble Alpes, financed by the Region Auvergne-Rhône-Alpes project SMART-PROTECT, supervised by Dr Vincent Acary and Dr Franck Bourrier, in collaboration with industrial partner Géolithe.

June 2015 – November 2019 Doctoral student at the University of Sydney and École nationale des ponts et chaussées, financed by an Australian Postgraduate Award and a University of Sydney Merit top-up scholarship.

Journal Articles

The influence of grain crushing and pore collapse on the formation of faults, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, in the Journal of Geophysical Research: Solid Earth, March 2026.
Codes. Data.

On the Moreau–Jean scheme with the Frémond impact law: energy conservation and dissipation properties for elastodynamics with contact, impact and friction, V. Acary & N.A. Collins-Craft, in the Journal of

Theoretical, Computational and Applied Mechanics, June 2025.
Codes. Data.

On the formulation and implementation of extrinsic cohesive zone models with contact, N.A. Collins-Craft, F. Bourrier & V. Acary, in Computer Methods in Applied Mechanics and Engineering, Volume 400, October 2022.
Codes. Data

A Cosserat Breakage Mechanics model for brittle granular media, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, in Journal of the Mechanics and Physics of Solids, Volume 141, August 2020.

Accepted Journal Articles

Incremental homogenisation of anisotropic porous media with evolving solid contact using a finite-element-based linear complementarity framework, L. Zhang, N.A. Collins-Craft & P. Braun.
Accepted at Computers & Geotechnics.

Accepted Book Chapters

Accounting for grain crushing and pore collapse for strain localization in rocks, N.A. Collins-Craft, J. Sulem, I. Stefanou & I. Einav, in Continuum Models and Discrete Systems, edited by P. Trovalusci.

Software

Hamersley
A software package written in Julia for the integration of higher-order and multi-physics in solid mechanical finite element simulations.

Julia Constitutive Model Integration
Codes that use the Julia language to integrate the model described in "A Cosserat Breakage Mechanics Model for Brittle Granular Media", as well as conduct linear stability analyses over given loading paths. Examples of model outputs and linear stability analyses are saved in a Zenodo deposit.

Preprints

On the formulation and implementation of mixed mode I and mode II extrinsic cohesive zone models with contact and friction, N.A. Collins-Craft & V. Acary.
Codes. Data.
Under review at Computer Methods in Applied Mechanics and Engineering.

Conference Presentations (Presenter Underlined)

A Fully Coupled Monolithic Finite Element Framework for Pore-Scale Hydromechanical Processes in Porous Media, L. Zhang, N.A. Collins-Craft & P. Braun, 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering, Munich, Germany 19/07/2026 – 24/07/2026.

Accounting for grain crushing and pore collapse for strain localization in rocks, N.A. Collins-Craft, J. Sulem, I. Stefanou & I. Einav, 15th International Symposium on Continuum Models and Discrete Systems, Rome, Italy 08/06/2026 – 12/06/2026.

Predicting the microstructural evolution of shear bands in crushable granular media, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, 8th International Conference on Coupled THMC Processes: Geomechanics for Energy and Environmental Applications, Nicosia, Cyprus, 20/10/2025 – 23/10/2025.

Micromechanical porous media simulation incorporating solid contact through enhanced Linear Complementarity Problem methodology, L. Zhang, N.A. Collins-Craft, J.-M. Pereira & P. Braun, 3rd International Conference on Energy Geotechnics, Paris, France, 17/06/2025 – 19/06/2025.

A new Moreau–Jean scheme that guarantees dissipativity, V. Acary & N.A. Collins-Craft, 11th Contact Mechanics International Symposium, Lyon, France 22/05/2024 – 24/05/2024.

Extending non-smooth contact mechanics to cohesive zone modelling, N.A. Collins-Craft, F. Bourrier & V. Acary, 7th International Conference on Computational Contact Mechanics, Turin, Italy, 05/07/2023 – 07/07/2023.

A non-smooth extrinsic cohesive zone model including contact and friction, N.A. Collins-Craft, F. Bourrier & V. Acary, The Seventh International Conference on Computational Modelling of Fracture and Failure of Materials and Structures, Prague, Czechia, 21/06/2023 – 23/06/2023.

A non-smooth cohesive zone model for rock fracture and contact, N.A. Collins-Craft, F. Bourrier, J. Gaume & V. Acary, European Geophysical Union General Assembly 2023, Vienna, Austria, 23/04/2023 – 28/04/2023.

A non-smooth formulation of an extrinsic cohesive zone model with contact and friction, N.A. Collins-Craft, F. Bourrier & V.Acary, 11th European Solid Mechanics Conference, Galway, Ireland, 04/07/2022 – 08/07/2022.

An extrinsic cohesive zone model with contact and friction, N.A. Collins-Craft, F. Bourrier & V.Acary, 18th European Mechanics of Materials Conference, Oxford, United Kingdom, 04/04/2022 – 06/04/2022.

An extrinsic cohesive zone model with contact developed in the non-smooth mechanics framework, N.A. Collins-Craft, F. Bourrier & V. Acary, 33rd Nordic Seminar on Computational Mechanics, Jönköping, Sweden, 25/11/2021 – 26/11/2021.

Unifying Breakage Mechanics with the Cosserat continuum to predict shear band localisation, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, International Congress of Theoretical and Applied Mechanics 2020+1, Milan, Italy, 22/08/2021 – 27/08/2021.

A theory to predict shear band formation in granular media with evolving grain size distributions, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Micro2Macro 2018, Reggio Calabria, Italy, 29/05/2018 – 01/06/2018.

A formulation of Breakage mechanics in the Cosserat continuum to predict shear band formation in granular media, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, 12th EURO-conference on Rock Physics & Geomechanics, Ma'ale HaHamisha, Israel, 05/11/2017 – 10/11/2017.

Conference Posters

The effect of grain size reduction on shear band formation, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Patterns in Geomechanics, Sydney, Australia, 29/01/2019 – 01/02/2019.

The effect of grain size reduction during shearing of granulated fault gouge, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, AGU Fall meeting, Washington D.C., United States of America, 10/12/2018 – 14/12/2018.

Modelling crushable granular media in the Cosserat continuum, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Navier Seminar, Paris, France, 05/03/2018.

Progress towards a constitutive model for crushable granular media using Cosserat continuum modelling, N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, ERC REALISM Kick-off Meeting, Paris, France, 15/03/2017.

Invited Seminars

Faults, friction, fracture and the modelling of pathologies in solid mechanics, Applied Mathematics seminar, Murdoch University, 13/01/2026.

Nonsmooth mechanics for fracture and friction problems in geomaterials, CivMin seminar, University of Toronto, 14/07/2025.

Nonsmooth mechanics for fracture and friction problems in geomaterials, SPREE seminar, Northwestern University, 28/02/2024.

La perspective Moreau-iste sur la fracture : extension de la mécanique nonsmooth aux modèles de zones cohésives extrinsèques avec contact et friction, laboratory seminar, Université de Montpellier, 18/01/2024.

Non-smooth mechanics for fracture propagation in geomaterials, ENAC Seminar Series, École Polytechnique Fédérale de Lausanne, 16/05/2023.

La mécanique non-lisse pour la modélisation de la fissuration : principes, premiers résultats et perspectives, MISES seminar, Sorbonne Université, 08/03/2023.

Formulating and implementing extrinsic cohesive zone models in the non-smooth mechanics framework, CERMES seminar, École nationale des ponts et chaussées, 22/10/2022.

Predicting strain localisation in crushable granular media using the Cosserat continuum, TRIPOP seminar, Inria Grenoble-Rhône-Alpes, 07/12/2020.

A Cosserat Breakage Mechanics model for brittle granular media and An Introduction to Julia, CoQuake seminar, École centrale de Nantes, 02/06/2020.

The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media, Granular Forum, the University of Sydney, 20/06/2018.

The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media, CERMES seminar, École nationale des ponts et chaussées, 19/02/2018.

The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media, ENS seminar, École normale supérieure de Paris, 03/03/2017.

Contributions to Software Primarily Authored by Others

Numerical Geolab
Software primarily authored by Ioannis Stefanou and Alexandros Stathas. Codes that provide a layer over the FEniCS finite element software to facilitate its usage for nonlinear, multiphysical and higher-order continuum mechanics problems. I am currently the active maintainer of the repository.

Students Supervised

2024 – Chloé Gergely, masters student in the program *Master Mathématiques et applications* of the *Université Claude Bernard Lyon 1*, co-supervised with Franck Bourrier & Vincent Acary. Continued on to a PhD in the group.

Teaching Duties

2026 Lecturer and Course Coordinator for MAS162 Foundations of Discrete Mathematics, a first year mathematics course at Murdoch University.

2026 Lecturer for USMA1Q *Méthodes Numériques* (Numerical Methods), a second year of engineering degree (fourth year of tertiary study) course at the Conservatoire national des arts et métiers.

Seminars to Masters Students

L'effet de l'evolution de l'échelle de longeur micro-structurel sur le comportement macroscopique constitutif des milieux granulaires, MSROE seminar, École nationale des ponts et chaussées, 11/01/2018. Approximately 30 students.

The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media, MSROE seminar, École nationale des ponts et chaussées, 05/01/2017. Approximately 30 students.

Seminars to Undergraduate Students

Modélisation du contact, de l'impact, du frottement et de la fracture, "Métiers de la recherche" seminar, École nationale des ponts et chaussées, 24/01/2025. Two groups of ten students each.

Working Papers

A novel multiscale method and its application to faults
N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav.

Shear band formation in the Saint-Maximin Limestone modelled with Breakage Mechanics
N.A. Collins-Craft, C. Doré-Ossipyan & J. Sulem.

Thermo-hydro-mechanical Cosserat Breakage Mechanics model
N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav.

Monolithic hydromechanical model for fluid flow and solid deformation
L. Zhang, N.A. Collins-Craft & P. Braun.

Comparison of the predictions of a fracture model with experimental results obtained on limestone samples
N.A. Collins-Craft & C. Silvani.

Prizes, Scholarships and other Academic Achievements

2022 Marie Skłodowska-Curie Actions Postdoctoral Award.

2017 Early Career Researcher grant to attend the 12th Euro-conference on Rock Physics & Geomechanics.

2016 Postgraduate Research Support Scheme.

2015 Australian Postgraduate Award, University of Sydney Merit Award (for three and a half years each).

2014 D.G. Walkom Prize for First Class Honours in Civil Engineering, Dean's List of Excellence in Academic Performance.

2013 D. Campbell-Allen Prize in Civil Engineering, invitation to participate in the Summer Research Scholarship programme.

2012 University of Sydney Academic Merit Prize, Dean's List of Excellence in Academic Performance, invitation to participate in the Summer Research Scholarship programme.

2011 University of Sydney Academic Merit Prize, Dean's List of Excellence in Academic Performance, invitation to participate in the Summer Research Scholarship programme.

2010 Awarded Roads and Maritime Services Scholarship in Civil Engineering for length of Bachelor's Degree.

Academic Service

2020 – Present Reviewer for Acta Geotechnica, Computer Methods in Applied Mechanics and Engineering, Computational Mechanics, Computers and Geotechnics, International Journal of Solids and Structures, Interpore Journal, Journal of Geophysical Research: Solid Earth, Mathematics in Engineering, Powder Technology, Rock Mechanics and Rock Engineering, and Scientific Reports.

2026 – Present Lecturer for the Western Australia Maths School, an outreach activity targeted at encouraging highly-engaged and high-performing students in Years 10–12 to study mathematics at university.

2026 – Present Committee Member for the Western Australia Junior Olympiad , an outreach activity targeted at Years 7–9 students with the aim of identifying students suitable to compete in the Australian Mathematical Olympiad.

2025 Organisational assistant for the 3rd International Conference on Energy Geotechnics, Paris, France, 17/06/2025 – 19/06/2025

2017 Organisational assistant for 6th International Conference on Coupled THMC Processes in Geosystems, Paris, France, 05/07/2017 – 07/07/2017.

Other Academic Qualifications

Qualification aux fonctions de Maître de Conférences in sections 36 and 60 of the Conseil National des Universités, 2022 qualification campaign.

Other Academic Work

2025 Tutor for *Programme Étudiants Réfugiés* (Refugee Students Program), one-to-one mathematics and informatics tutoring of refugee students to help bring them to the required level for further study in France.

2023 Tutor for MAP101 *Analyse élémentaire et introduction au calcul scientifique* (Elementary real analysis and introduction to scientific computing), first year applied mathematics and informatics course at Université Grenoble Alpes.

2019 Assignment marking and substitute tutor for CIVL2410 Soil Mechanics, second year civil engineering course at the University of Sydney.

2017 – 2018 Scientific English editing of a wide variety of scientific papers written by other ENPC researchers.

2016 Tutor for ENGG1801 Engineering Computing, first year general engineering course at the University of Sydney.

2015 Tutor for CIVL3411 Geotechnical Engineering, third year civil engineering course at the University of Sydney.

Non-Academic Employment

2017 – 2018 English tutor. Self-employed English tutor in France.

2015 – 2016 Kambala. Debating coach for the Independent Schools Debating Association and Archdale Schools competitions.

2011 – 2015 The Arts Unit. Debating adjudicator for public primary and high schools in the Northern Sydney region.

2015 Maths and Physics Tutor. Self-employed private tutor for first year Maths and Physics subjects.

2011 – 2014 Roads and Maritime Services. Three separate stints over the summer break in association with my scholarship. I played a role in project management of a A$400 million public works (Northern Beaches Hospital Connectivity and Network Enhancement) in the Project Development unit (13 – 14), developer liaison on large projects such as Wynyard Walk and Moorebank Units Relocation (Holsworthy Barracks) in the Project Management unit (12 – 13), and material behaviour research in the Pavements unit (11 – 12).

2011 – 2014 Trombone Tutor. Self-employed.

2011 Engineering Studies Tutor. Self-employed.

Languages

English (mother tongue), French (proficient).

Contact Information

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